



Storage / Refinery
VRU Engineering Questionnaire

1. Portfolio overview

Combined figures for all relevant storage and refinery sites.

Number of relevant sites

Country

Storage, tank farm, and refinery sites requiring VRU review

City / region

Typical inventory across all sites (t)

Normal operating inventory

Total usable capacity across all sites (t)

Total monthly throughput across all sites (t/month)

Maximum safe working inventory

Monthly product received, transferred, or dispatched

2. Proposed VRU project

Provide the scope and contacts for the proposed VRU package, not the wider refinery programme.

Proposed VRU project name / reference

Site address / area reference

Country

City / region

Technical contact

Contact email

Expected VRU project start

Target VRU commissioning date

3. Facility and transfer sources

Select every source that can feed tanks connected to this vapor system. Complete applicable fields only.

Facility type (dropdown - select one)



Truck receipt

Refinery / pipeline transfer

Rail receipt

Marine / barge receipt

Truck receipt frequency (events/day, if applicable)

Simultaneous trucks (if applicable)

Pipeline/refinery transfer flow (m3/h, if applicable)

Simultaneous transfer lines

Normal / peak transfer hours per day

Transfer pressure (barg or kPa)

Tank receipt / fill method (dropdown - select one)



Submerged describes discharge below the liquid level; bottom fill describes entry through a lower tank nozzle.

4. Simultaneous operations and design case

Use the absolute worst credible case, not monthly throughput or an average operating day.

Maximum simultaneous loading arms / hoses

Maximum simultaneous truck bays

Maximum simultaneous marine berths

Maximum simultaneous refinery / pipeline lines

Average active transfer rate (m3/h per connection)

Peak active transfer rate (m3/h per connection)

Worst-case simultaneous scenario

5. Tank inventory



List tanks connected to the proposed VRU vapor collection system. Add rows on a continuation sheet if needed.

Tank ID	Product	Tank type	Normal inventory (t)	Usable capacity (t)	Throughput (t/month)	VRU connected?
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Tanks gas blanketed - Yes

No

Blanketing gas and set pressure (if applicable)

6. Vapor and product characteristics

Use laboratory or current design-basis data where available.

Headspace vapor composition / GC analysis

Vapor molecular weight (g/mol)

Vapor lower explosive limit (% by vol)

Water saturated vapor stream? (dropdown - select one)



Corrosive / toxic components

Additional vapor or product notes

7. Vapor system and piping

Complete available values. Mark not available in the notes when design data is pending.

Vapor header nominal size (mm)

Total vapor piping length (m)

Tank P/V vent positive setting (mbar or Pa)

Tank P/V vent vacuum setting (mbar or Pa)

Existing flame/detonation arrestor - Yes

No

Vapor piping material

Piping constraints or notes

Vapor-header tie-in to VRU plot distance (m)

Tie-in to VRU plot elevation change (m)

8. Utilities and hazardous area

Use IECEx/ATEX Zone terminology unless another owner/project standard applies.

Available voltage (V)

Electrical phase (dropdown - select one)



Electrical frequency (dropdown - select one)



Hazardous-area classification (dropdown - select one)



Gas group / certification requirement

Instrument air pressure (bar)

Cooling water: flow, temperature, pressure

Installation-site internet access (dropdown - select one)



9. Installation, safety, and compliance

Complete applicable fields and attach owner specifications, drawings, and permits where available.

VRU installation space constraints

Available construction / lifting equipment

Applicable environmental permit or emissions standard

Required VOC recovery efficiency (%)

Noise limit (dBA and distance)

Owner / project standards (IEC, ISO, EN, API if required)

Usable VRU plot length (m)

Usable VRU plot width (m)

Plot dimensions, boundaries, access, setbacks, height, or layout restrictions

10. Controls, emissions, and guarantees

Capture the owner acceptance requirements and control-room integration basis before final VRU scoping.

Control-system interface (dropdown - select one)



Existing DCS / SCADA vendor

Local PLC requirement (dropdown - select one)



Required control-room handshake (dropdown - select one)



CEMS / analyzer required? (dropdown - select one)



Guaranteed outlet limit (mg/Nm³ or ppmv)

Outlet pollutant / guarantee basis and test method

11. Refinery unit design basis

Complete if the proposed VRU serves refinery process equipment or refinery transfer systems.

Refinery unit / process area

Connected vapor sources

Operating mode (dropdown - select one)



Expected operating hours per year

Normal vapor flow (Nm3/h)

Design vapor flow (Nm3/h)

VRU inlet pressure and temperature

Downstream destination and pressure

Refinery design-basis notes

12. Reporting, attachments, and sign-off

Check documents being supplied with the completed PDF. Supporting files must be sent separately.

P&ID / piping drawings

GC or laboratory analysis

Plot plan / general arrangement

Tank datasheets

Photographs

Owner / EPC specifications

Environmental permit

Other supporting document

Required reports, approvals, and deliverables

Other VRU requirements or constraints

Prepared by

Date